

Stretch GazeBot — Reproduce in 1 Page

Control a Hello Robot Stretch 3 with your head using Xreal One Pro AR glasses. Project: `~/lab/life_home/xreal_head_control`

1. You need

- Xreal One Pro + USB-C cable (data)
- Hello Robot Stretch 3 reachable on LAN (default 172.22.243.8)
- Linux PC, Python 3.7+

2. PC — one-time setup

Clone repo, install deps, enable glasses ethernet (Settings → Developer → Ethernet).

```
git clone <repo> ~/lab/life_home/xreal_head_control
cd ~/lab/life_home/xreal_head_control
pip install -r requirements.txt
ping 169.254.2.1 # must reply
```

3. Robot — start two SSH terminals

`scp stretch_relay.py` to the robot once:

```
scp stretch_relay.py
hello-robot@172.22.243.8:~/het_ros2_ws/
```

Terminal A (driver):

```
ssh hello-robot@172.22.243.8 # pw: <ask lab admin>
conda deactivate
ros2 launch stretch_core stretch_driver.launch.py
```

Terminal B (relay):

```
ssh hello-robot@172.22.243.8
conda deactivate
python3 ~/het_ros2_ws/stretch_relay.py
# wait for: "Waiting for connection on port 9090..."
```

4. PC — run the bridge

```
cd ~/lab/life_home/xreal_head_control
python3 stretch_head_control.py
```

Hold the glasses still ~0.5 s for calibration → put them on → press **T** to zero your view. Move your head, robot follows.

5. Keys

M toggle LOOK/DRIVE • **SPACE** hold to drive (DRIVE) • **T** zero view • **R** recalibrate • **L** lock • **Q** quit

6. Tune (top of `stretch_head_control.py`)

```
STRETCH_IP = "172.22.243.8" # robot IP
HEAD_PAN_SCALE = 0.025 # ↑ = more sensitive L/R
HEAD_TILT_SCALE = 0.015 # ↑ = more sensitive U/D
# Wrong direction? flip sign on line 163 or 164
```

7. Common fixes

- Connection refused → relay not running on robot.
- Trajectory server not found → driver not running, or ``ros2 action list`` shows different name.
- `rcipy` import error → run ``conda deactivate`` on the robot first.
- Cannot ping 169.254.2.1 → enable ethernet in glasses dev menu / try another USB-C port.
- Drift → press R, hold still.

8. Files at a glance

PC: `stretch_head_control.py` · `src/imu_reader_fixed.py` · `src/head_tracker.py`

Robot: `stretch_relay.py` (in `~/het_ros2_ws/`)

Test: `imu_server.py` + `imu_viz.html` (3D cube in browser)

Architecture: **[Glasses]** → USB-C TCP → **[PC bridge]** → TCP JSON :9090 → **[Robot relay]** → ROS 2 action → head pan/tilt joints. See [Stretch_GazeBot_Reproduction_Guide.pdf](#) for full detail.